# ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY

Surampalem, Kakinada, Andhra Pradesh

A Project Report on

## DESIGN THINKING FOR AI-POWERED USED CAR PRICE PREDICTOR – AUTOVAL AI

Submitted in partial fulfillment of the requirements for the course

**Design Thinking**




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**Academic Year: 2025 – 2026**

# CERTIFICATE

This is to certify that the project titled **"Design Thinking for AI Powered Used Car Price Predictor – AUTOVAL AI"** has been successfully completed and submitted by the following students of the Department of Computer Science and Engineering, Aditya College of Engineering and Technology, Surampalem, Kakinada, Andhra Pradesh, in partial fulfillment of the requirements of the **Design Thinking** course during the academic year **2025–2026**.

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The students have demonstrated a thorough understanding of the Design Thinking methodology and have applied it effectively to address the problem of healthcare accessibility in rural and low-literacy communities. The project work presented in this report is genuine and has been carried out under proper academic guidance.

| | |
|---|---|
| **Faculty Signature** | **Head of Department** |
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# ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who contributed to the successful completion of this Design Thinking project titled "Design Thinking for AI Powered used Car Price Predictor – AUTOVAL AI."

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– Team AUTOVAL AI

# ABSTRACT

Access to accurate and transparent pricing in the used car market remains a significant challenge for many individuals across India. Car sellers and buyers often rely on guesswork, dealer influence, or inconsistent online listings, leading to unfair pricing, lack of trust, and inefficient decision-making. Factors such as vehicle condition, mileage, age, brand value, and fuel type further complicate price estimation, especially for first-time sellers who lack domain knowledge.

This project applies a data-driven and user-centric approach using Machine Learning techniques to address the problem of unreliable car price estimation. Through systematic stages of problem understanding, data collection, model development, and evaluation, our team developed **AutoVal-AI**, an AI-powered car price prediction system designed to provide accurate and instant valuation of used vehicles.

AutoVal-AI incorporates several key features aimed at improving transparency and ease of use: real-time price prediction using a trained regression model, analysis based on essential inputs such as company, model, year, kilometres driven, and fuel type, a confidence-based price range estimation, depreciation insights over time, and identification of similar car listings for comparison. The system is built as a full-stack web application using Flask for backend processing and an interactive frontend interface for user engagement.

The predictive model is trained on real-world used car data sourced from Indian market listings, enabling it to capture practical pricing trends and patterns. Among multiple models evaluated, Gradient Boosting demonstrated superior performance in terms of accuracy and error reduction, making it the preferred choice for deployment.

The application prototype emphasizes simplicity, responsiveness, and accessibility, allowing users to obtain reliable price estimates within seconds without requiring technical expertise. Testing and evaluation of the system indicate that the model provides reasonably accurate predictions and meaningful insights, helping users make informed buying and selling decisions.

This report documents the complete development process of the project, including dataset preparation, model selection, system design, implementation, and performance evaluation, demonstrating how Artificial Intelligence can enhance transparency and efficiency in the used car market.

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# 1. INTRODUCTION

## 1.1 Design Thinking: An Overview

Design Thinking is a non-linear, iterative problem-solving methodology that focuses on understanding user needs, redefining complex problems, and creating innovative, practical solutions through continuous prototyping and testing. Popularized by institutions such as the Stanford d.school and design firms like IDEO, Design Thinking has become an essential framework in modern engineering and technology development.

Unlike traditional approaches that begin with a predefined technical solution, Design Thinking starts with empathy—understanding the real challenges faced by users. This approach ensures that solutions are not only technically sound but also meaningful, usable, and relevant in real-world scenarios. By focusing on user experience and iterative improvement, Design Thinking enables the development of systems that effectively solve practical problems.

In the context of this project, Design Thinking plays a crucial role in identifying the challenges faced by individuals in determining the correct resale value of used cars and developing a solution that is both accurate and user-friendly.

## 1.2 Importance of Human-Centered Design

Human-centered design places users at the core of the entire development process. In applications related to financial decision-making—such as buying or selling vehicles—this approach is particularly important because users often rely on the system's output to make critical economic decisions.

In the used car market, users include first-time sellers, buyers, and individuals with limited knowledge of market trends. These users require a system that is simple, transparent, and trustworthy. A technically advanced model is ineffective if users cannot understand how to interact with it or interpret its results.

Human-centered design principles emphasize simplicity, clarity, and accessibility. This involves creating intuitive interfaces, minimizing complex inputs, and presenting results in an understandable format. In this project, the system is designed to ensure that even users with minimal technical knowledge can easily input car details and obtain accurate price predictions without confusion.

## 1.3 Challenges in Used Car Price Estimation

The used car market in India is vast and highly unorganized, making price estimation a complex task. Several challenges contribute to the lack of transparency and reliability in determining the resale value of vehicles:

- **Lack of pricing transparency:** Car prices vary significantly across different platforms and dealers, leading to confusion among buyers and sellers.
- **Dependence on middlemen:** Many individuals rely on dealers or brokers who may manipulate prices for profit, resulting in unfair transactions.
- **Multiple influencing factors:** Car valuation depends on several parameters

such as brand, model, year of manufacture, fuel type, and kilometres driven, making manual estimation difficult.

- **Inconsistent online listings:** Prices listed on websites may not accurately reflect actual market value due to negotiation margins or incomplete data.
- **Limited awareness:** First-time sellers often lack knowledge about depreciation trends and market demand, leading to underpricing or overpricing of vehicles.

## 1.4 Digital and Data Challenges

Although digital platforms for buying and selling cars have grown rapidly, they often fail to provide reliable and personalized price estimation. Most platforms display static listings instead of dynamic predictions based on user-specific inputs.

Additionally, the accuracy of price prediction depends heavily on the quality and diversity of data. Without proper data preprocessing and model selection, predictions can be misleading. Users also face challenges in understanding complex pricing factors and interpreting large datasets.

These challenges highlight the need for an intelligent, data-driven system that can process multiple variables and generate accurate, real-time predictions in a user-friendly manner.

## 1.5 The AutoVal-AI Solution

To address these challenges, our team developed **AutoVal-AI**, an AI-powered car price prediction system designed to provide accurate and instant valuation of used vehicles.

AutoVal-AI leverages Machine Learning techniques to analyze key features such as car company, model, year of purchase, fuel type, and kilometres driven to predict the market price. The system is built as a full-stack web application using Flask for backend processing and an interactive frontend interface for seamless user interaction.

The application integrates several important features, including real-time price prediction, confidence-based price range estimation, visualization of depreciation trends, and insights based on vehicle usage patterns. The predictive model is trained on real-world car listing data, ensuring that the results reflect actual market conditions. Among various models tested, Gradient Boosting provided the best performance in terms of accuracy and error minimization.

The system is designed with simplicity and usability in mind, enabling users to obtain reliable price estimates within seconds. By reducing dependency on intermediaries and improving transparency, AutoVal-AI empowers users to make informed decisions while buying or selling used cars.

# 2. EMPATHIZE STAGE

The Empathize stage forms the foundation of the Design Thinking process. During this phase, our team focused on understanding the experiences, behaviors, and challenges faced by individuals involved in buying and selling used cars. The objective was to gain a deep understanding of user needs, frustrations, and expectations related to car price estimation, rather than relying on assumptions.

Our target users included first-time car sellers, buyers, car owners, and individuals who depend on online platforms or dealers to determine the value of their vehicles. By engaging directly with these users, we aimed to identify real-world problems and uncover gaps in the current system of car valuation.

## 2.1 Research Methods

To gather meaningful insights, our team used a combination of primary and secondary research methods:

**Interviews and Conversations:** We conducted informal discussions with car owners, students, and individuals who had recently bought or sold used cars. These conversations highlighted common issues such as confusion about pricing, dependence on dealers, and lack of trust in online listings. Many users expressed uncertainty about whether they were getting a fair deal.

**Observation:** We observed how users interact with existing car-selling platforms and price estimation tools. It was noted that users often browse multiple websites, compare prices manually, and still remain unsure about the correct value. Many users rely on approximate guesses rather than data-driven decisions.

**Online Research:** We analyzed data from online car marketplaces and studied general pricing trends in the Indian used car market. This helped us understand how factors like car age, mileage, fuel type, and brand influence pricing. Secondary research also validated the issues identified during interviews and observations.

## 2.2 Key Findings

Through our empathy research, we identified the following critical pain points experienced by users in rural and low-literacy communities:

- **Lack of price transparency:** Users find it difficult to determine the true market value of a car due to inconsistent pricing across platforms.
- **Dependence on dealers and middlemen:** Many users rely on brokers or dealers who may undervalue or overprice vehicles for their own benefit.
- **Difficulty in considering multiple factors:** Car price depends on various parameters such as company, model, year, fuel type, and kilometres driven. Users struggle to combine all these factors to estimate an accurate price.
- **Uncertainty and lack of confidence:** Users often feel unsure whether the price they are setting or accepting is fair, leading to hesitation in decision-making.

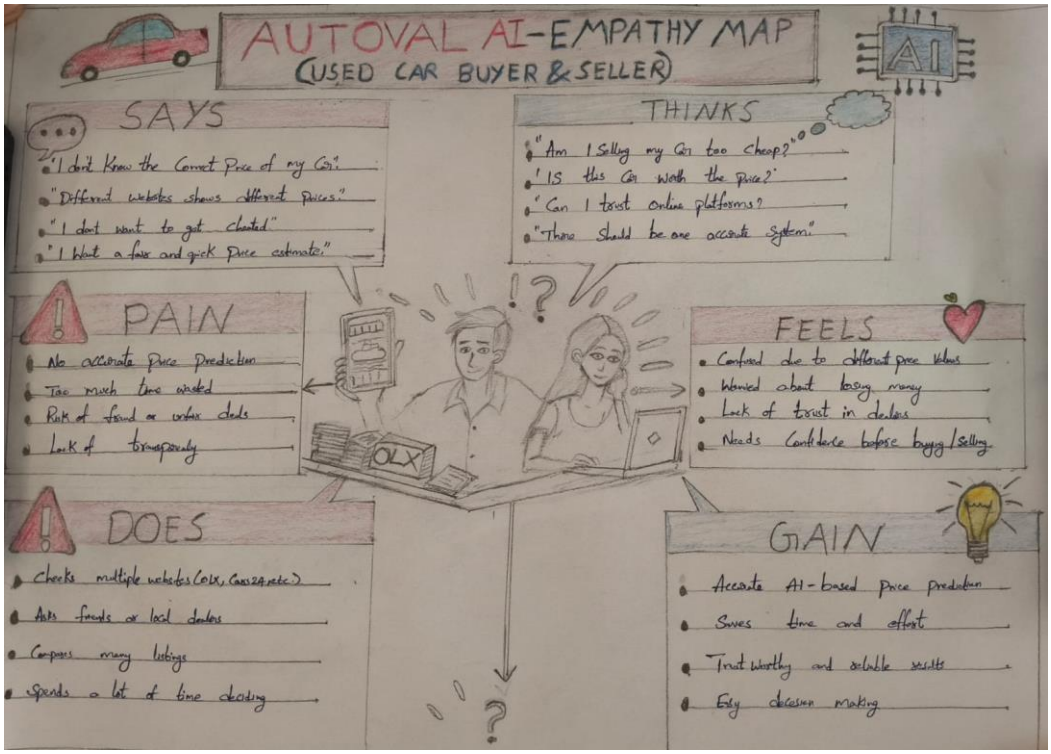
- **Time-consuming process:** Comparing multiple listings, analyzing trends, and negotiating prices takes significant time and effort.

## 2.3 Empathy Maps

Based on the insights gathered, we developed empathy maps to better understand user behavior and mindset. These maps captured what users say, think, feel, and do during the process of buying or selling a used car.

A recurring pattern observed was the feeling of confusion and lack of trust in the system. Users expressed concerns about being misled or making poor financial decisions. At the same time, there was a strong need for a simple, reliable, and quick solution that could provide accurate price estimates without requiring technical knowledge

. This deep understanding of user challenges helped us clearly identify the problem space and guided the development of a solution that focuses on accuracy, transparency, and ease of use.



# 3. DEFINE STAGE

The Define stage involves synthesizing the insights gathered during the Empathize phase to clearly articulate the core problem. This stage transforms user observations, challenges, and feedback into a focused problem statement that guides the ideation and development of an effective solution.

## 3.1 Problem Statement:

Many individuals in the used car market struggle to determine the accurate resale value of vehicles due to lack of price transparency, dependence on dealers, and difficulty in analyzing multiple influencing factors. There is a need for a reliable, data-driven, and user-friendly system that can provide instant and accurate car price predictions, enabling users to make informed decisions without relying on intermediaries.

## 3.2 Analysis of the Problem

Our analysis revealed that the issue of car price estimation is not simply a lack of tools, but a broader problem involving trust, complexity, and accessibility. Existing platforms provide listings but fail to offer personalized and accurate predictions based on user-specific inputs.

Most current solutions assume that users have sufficient knowledge about market trends, depreciation, and pricing factors. However, in reality, many users—especially first-time sellers—lack this understanding. As a result, they either undervalue their vehicles or depend heavily on dealers and brokers who may not always provide fair pricing.

The core need is not just a platform that displays prices, but a smart system that simplifies decision-making. The solution must focus on accuracy, transparency, and ease of use. It should be capable of analyzing multiple parameters such as car brand, model, year, fuel type, and kilometres driven, and convert this complex data into a simple and understandable price estimate.

Additionally, the system must work efficiently in real-world conditions where users expect quick results, minimal input effort, and clear output without technical complexity communities.

## 3.3 User Personas

To further refine our understanding, we developed representative user personas based on our research:

**Persona 1 – Ravi (Age 35, First-Time Car Seller):** Ravi wants to sell his car but has no idea about its current market value. He checks multiple websites but finds different prices everywhere, leaving him confused. He needs a reliable way to estimate a fair price quickly.

**Persona 2 – Anitha (Age 24, Buyer):** Anitha is planning to buy a used car within her budget. She wants to ensure that the price quoted by sellers is reasonable. She needs a tool that helps her verify whether a car is

overpriced or fairly valued.

**Persona 3 – Arjun (Age 40, Car Owner):**

Arjun wants to upgrade his car and is curious about its resale value. He prefers a fast and simple solution where he can enter basic details and instantly get an estimate without consulting a dealer.

## 3.4 Design Requirements

Base Based on the problem analysis and user personas, we identified the following key design requirements for the solution:

- Simple and intuitive user interface for easy interaction
- Minimal input fields focusing only on essential parameters
- Real-time price prediction using Machine Learning
- Consideration of key factors such as company, model, year, fuel type, and kilometres driven
- Instant results with clear and understandable output
- Display of price range to reflect prediction uncertainty
- Data-driven accuracy based on real-world car listings
- Fast performance with minimal loading time
- Responsive design for both desktop and mobile users

# 4. IDEATE STAGE

The Ideate stage is where creativity is combined with structured thinking to explore possible solutions. During this phase, our team conducted collaborative brainstorming sessions to generate a wide range of ideas to solve the problem of inaccurate and non-transparent used car price estimation. The goal was to think broadly, encourage innovation, and explore multiple approaches before selecting the most effective solution.

## 4.1 Brainstorming Sessions

Our team conducted several brainstorming sessions using techniques such as mind mapping, "How Might We" questions, and rapid idea sketching. These sessions helped us focus on user needs identified in the Empathize and Define stages while encouraging creative thinking.

The following "How Might We" questions guided our ideation process:

- How might we help users accurately estimate the value of their car without expert knowledge?
- How might we reduce dependence on dealers and middlemen for price evaluation?
- How might we simplify the process of considering multiple factors affecting Car price?
- How might we provide instant results to save users time and effort?
- How might we make the system easy to use for users with minimal technical skills?

## 4.2 Ideas Generated

Through brainstorming, we generated several potential ideas and features:

1. **Manual Price Calculator:** A rule-based system where users input car details and receive a calculated estimate based on predefined formulas.
2. **Static Price Tables:** Providing average prices for cars based on brand and year, allowing users to compare values manually.
3. **Machine Learning-Based Prediction:** Develop a predictive model that learns from real-world data and provides accurate price estimation based on multiple factors.
4. **Price Range Estimation:** Displaying a minimum and maximum price range instead of a single fixed value to improve reliability.
5. **Depreciation Visualization:** Show how the car's value decreases over time using charts or graphs to help users understand pricing trends.
6. **Smart Insights and Condition Analysis:** Provide additional insights such as car age, usage patterns, and condition rating based on input data.
7. **Simple Web Interface:** Design a clean and user-friendly interface where users can easily input details and get results instantly.

### 4.3 Solution Selection

After generating multiple ideas, the team evaluated each concept based on three key criteria: feasibility (technical implementation), desirability (user needs), and viability (practical usefulness).

Through discussion and analysis, we decided to develop a **Machine Learning-based web application** that integrates the most valuable features into a single platform—**AutoVal-AI**. This approach was chosen because it provides higher accuracy compared to manual methods, reduces user effort, and delivers instant, data-driven results.

The final solution combines key features such as real-time price prediction, price range estimation, depreciation insights, and a simple user interface. This integrated approach ensures that users receive not only accurate predictions but also meaningful insights that support better decision-making.

# 5. PROTOTYPE STAGE

The Prototype stage transforms ideas into a functional and testable representation of the proposed solution. During this phase, our team focused on developing the **AutoVal-AI web application prototype**, ensuring that the system is not only technically functional but also user-friendly, responsive, and visually engaging.

The prototype integrates Machine Learning prediction with an interactive frontend interface, allowing users to input car details and instantly receive an estimated price. Special attention was given to interface design, navigation flow, and clarity of output to enhance user experience.

## 5.1 Design Goals

The prototype was developed with the following design goals:

- **Simplicity:** The interface is designed to be clean and intuitive, requiring minimal effort from users. Only essential input fields are included, reducing complexity and making the system easy to use even for first-time users.
- **Accuracy and Reliability:** The system uses a trained Machine Learning model to provide data-driven price predictions, ensuring that results are realistic and based on actual market trends.
- **Instant Feedback:** The application delivers results immediately after user input, eliminating delays and improving user satisfaction.
- **Clarity of Output:** The predicted price is displayed clearly along with a price range and supporting insights, helping users easily understand the results.
- **User Engagement:** Interactive elements such as sliders, dynamic updates, and visual indicators (like depreciation and condition insights) are included to enhance user interaction.

## 5.2 Design Approach

The team adopted a **web-based, responsive design approach**, ensuring that the application works smoothly on both desktop and mobile devices. The frontend was developed using HTML, CSS, and JavaScript, while the backend was implemented using Flask to handle user input and generate predictions.

The visual design follows a modern, dark-themed interface with highlighted accents to draw attention to key elements such as predicted price and inputs. The layout is structured to guide users step-by-step—from entering car details to viewing results—without confusion.

The system architecture includes:

- A user interface for input collection
- A backend server for processing requests
- A trained Machine Learning model for prediction
- A dataset containing real-world car information

## 6. APPLICATION PROTOTYPE SCREENS

The following figures present the high-fidelity user interface screens of the **AutoVal-AI web application**. Each screen is designed to provide a smooth, interactive, and user-friendly experience while enabling accurate car price prediction using Machine Learning. The interface follows a modern dark theme with highlighted elements to improve visibility, usability, and engagement.

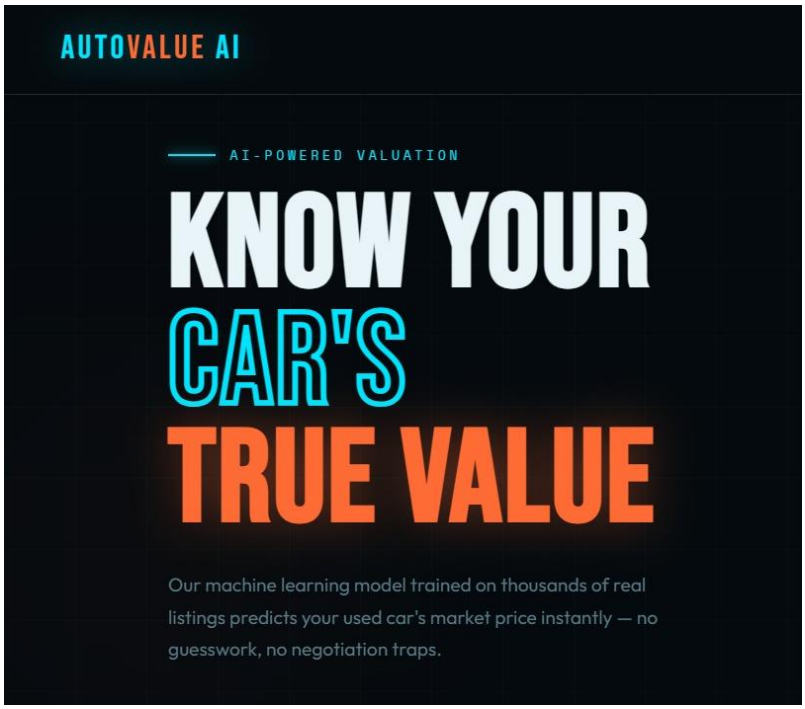



**Figure 1: Landing Page**

The Landing Page acts as the entry point of the AutoVal-AI application. It prominently displays the heading *"Know Your Car's True Value"* along with a short description explaining the purpose of the system. The visually appealing design and bold typography immediately capture user attention and communicate the AI-powered nature of the application.

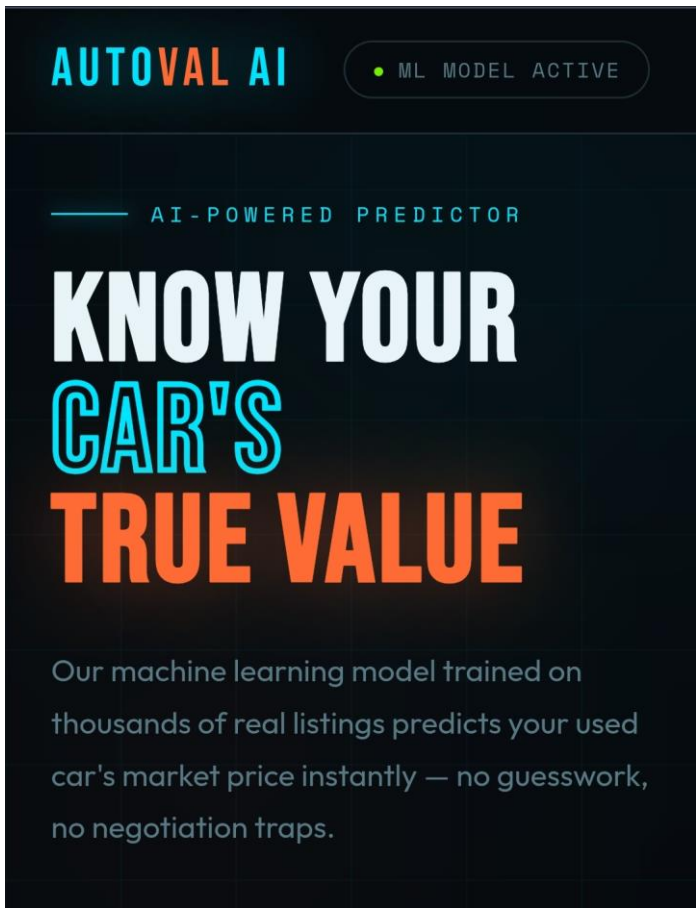



**Figure 2: Application Header**

The header contains the application branding “AutoVal AI” and a status indicator showing that the Machine Learning model is active. This consistent navigation bar enhances user trust and provides system status visibility.

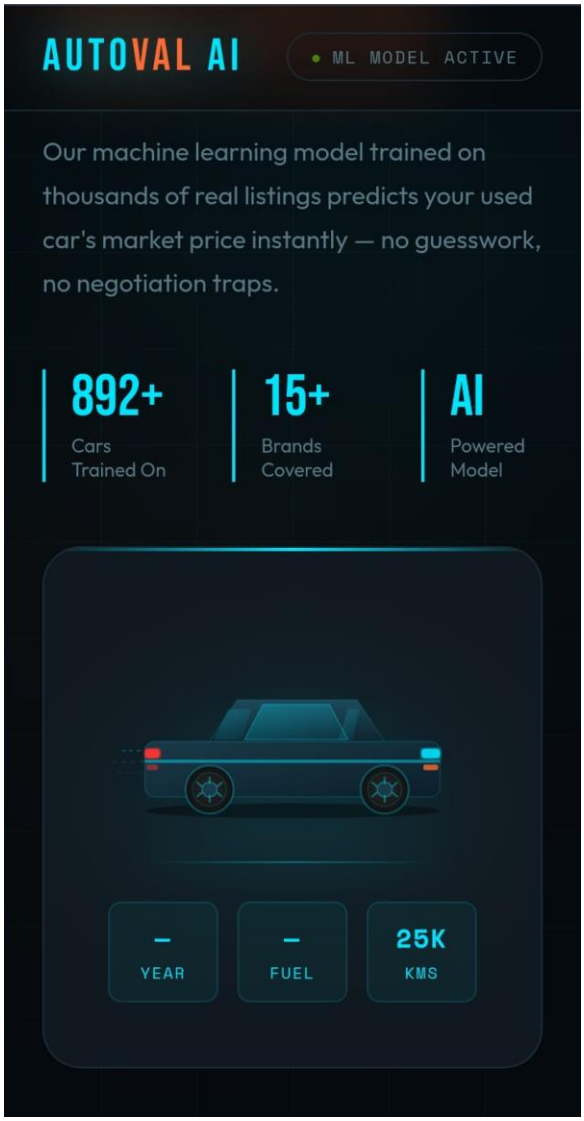



**Figure 3: Statistics & Model Highlights Section**

This section highlights important statistics such as the number of cars used for training, brands covered, and the AI-based prediction system. It helps users understand the reliability and scale of the model.

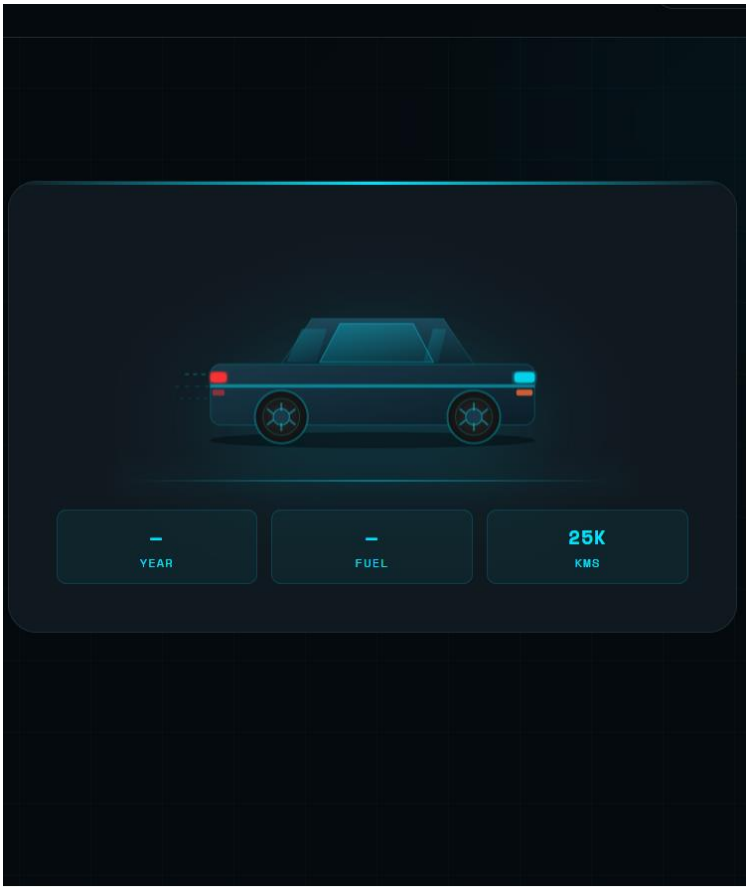



**Figure 4: Car Visual Display Panel**

This panel visually represents the car and dynamically displays key attributes such as year, fuel type, and kilometres driven. It enhances user interaction and provides a graphical summary of inputs.

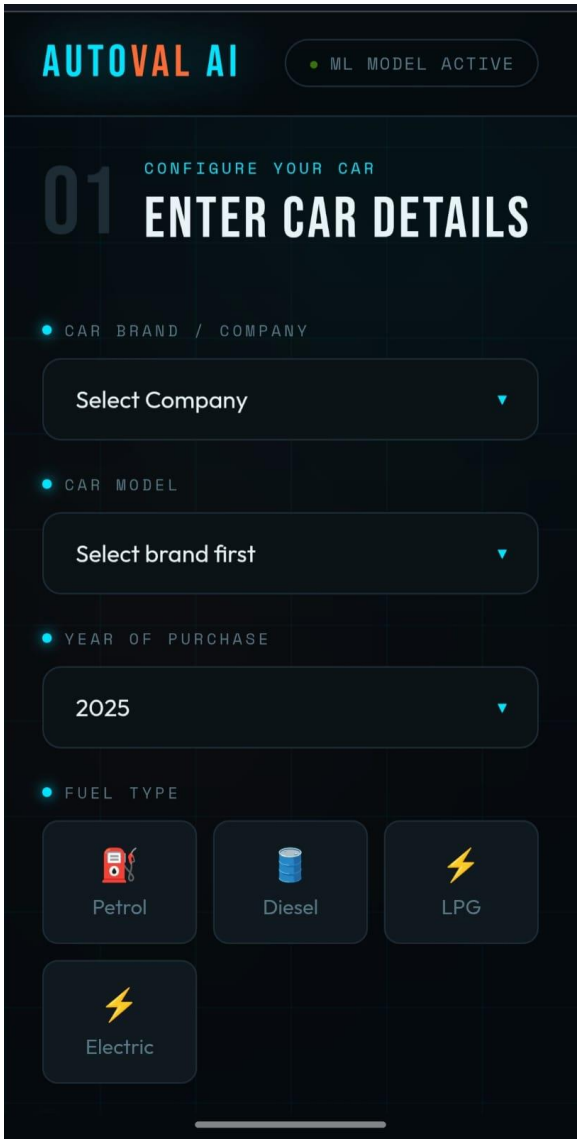



**Figure 5: Car Details Input Form**

This is the main input section where users enter car details such as brand, model, and year. The structured layout ensures clarity and ease of use, making it suitable even for first-time users.




**Figure 6: Fuel Type Selection Interface**

The fuel selection feature allows users to choose the type of fuel using visually distinct buttons and icons. This design makes selection intuitive and eliminates confusion.




**Figure 7: Year Selection Dropdown**

The year selection dropdown enables users to choose the manufacturing or purchase year easily. It avoids manual errors and ensures accurate input.




**Figure 8: Kilometres Driven Slider**

This feature allows users to input kilometres driven using an interactive slider. The real-time value display improves usability and provides instant feedback.

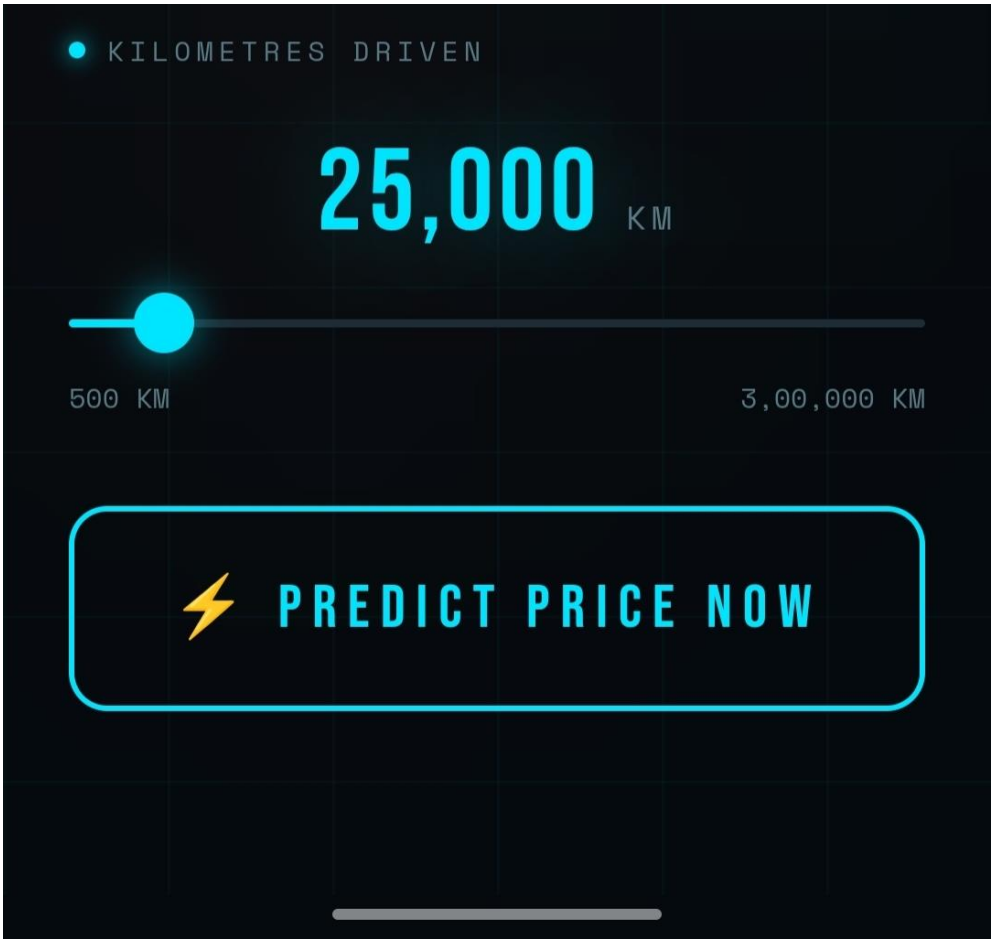



**Figure 9: Predict Price Button**

This is the main action button that triggers the Machine Learning model. The button is visually highlighted to attract attention and ensure easy interaction.

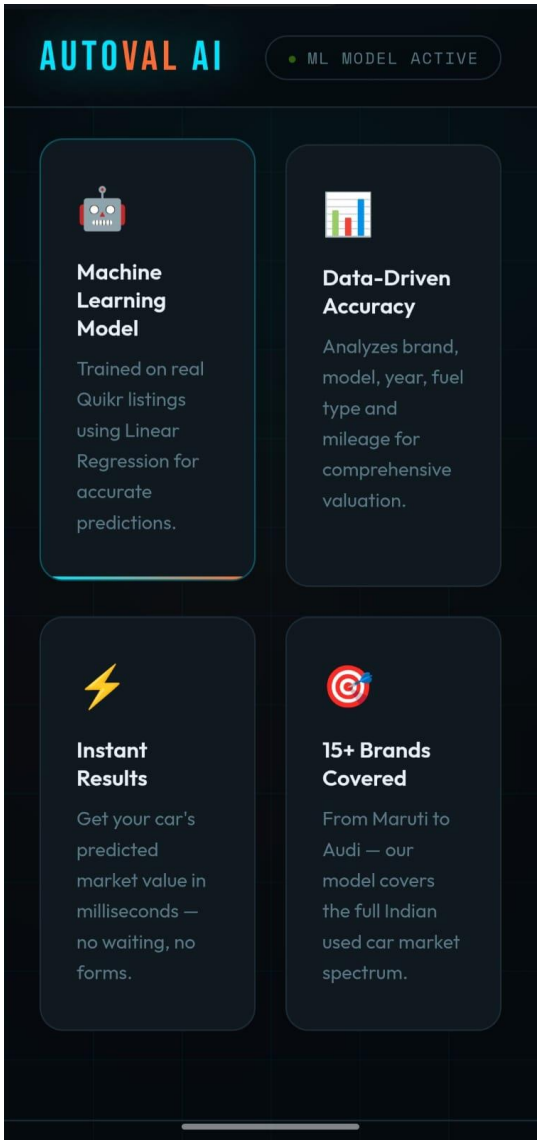



**Figure 10: Feature Highlights Section**

This section explains the core features of the system:

- Machine Learning model
- Data-driven accuracy
- Instant results
- Multiple brands support

It reinforces user trust and showcases system capabilities.

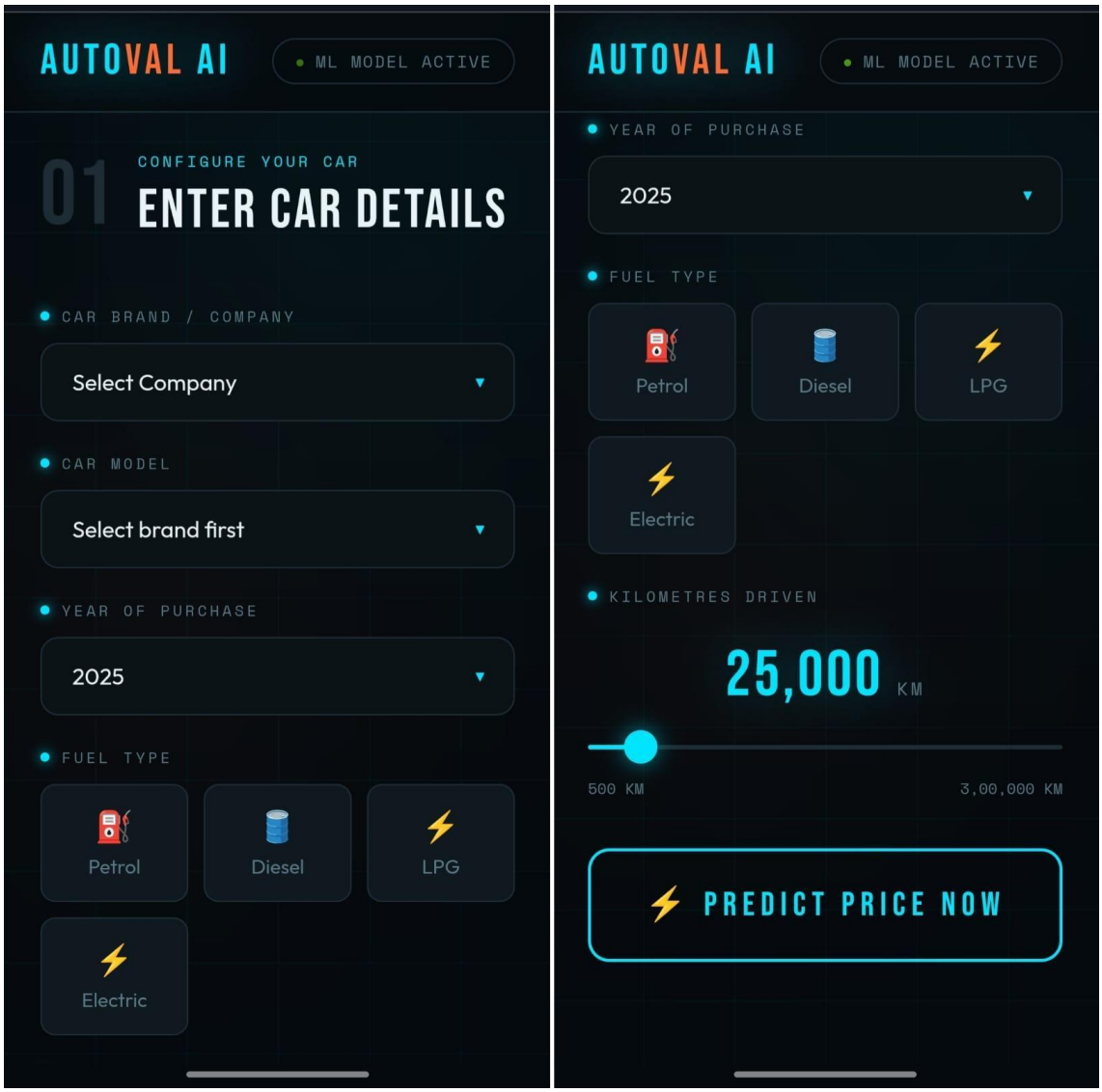



**Figure 11: Overall Application Interface**

This figure provides a complete overview of the application, showing how all components—landing page, input form, and features—are integrated into a single system. It demonstrates the smooth workflow from input to prediction.

# WORKING OF THE SYSTEM

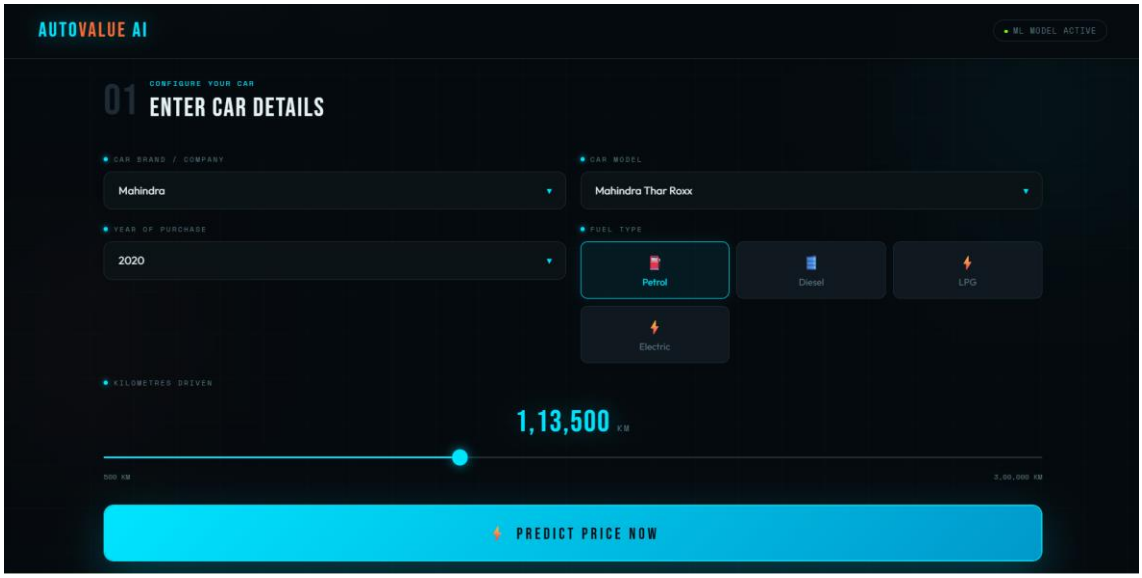



**Figure 12: Input Example for Prediction**

This figure shows an example where the user enters car details such as company, model, year, fuel type, and kilometres driven. The interface allows easy configuration using dropdowns and sliders. After entering the details, the user clicks the **"Predict Price Now"** button to generate the estimated price.




Figure 13: Predicted Price Output

This figure displays the predicted market value generated by the Machine Learning model. The system provides:

- Estimated price (₹10,46,471)
- Price range for reliability
- Confidence level (accuracy indicator)
- Car condition insights (Moderate mileage)

The output also includes visual elements such as:

- Depreciation graph (value over time)
- Price factor analysis (year, km, fuel, brand)
- Condition verdict

This helps users understand not only the price but also the factors influencing it.

# 7. TESTING STAGE

The Testing stage is the final phase of the Design Thinking process, where the developed prototype is evaluated by users to validate its effectiveness, usability, and accuracy. This stage helps in identifying whether the solution successfully addresses the problems defined earlier and highlights areas for improvement.

## 7.1 Testing Methodology

The AutoVal-AI web application was tested with a group of participants including fellow students, faculty members, and individuals who have experience in buying or selling used cars. The participants had varying levels of technical knowledge to ensure diverse feedback.

Users were asked to perform specific tasks such as:

- Entering car details (brand, model, year, fuel type, kilometres driven)
- Using the slider and dropdown inputs
- Clicking the "Predict Price Now" button
- Observing and interpreting the predicted price

Participants were also encouraged to provide feedback on ease of use, design clarity, and usefulness of the system.

## 7.2 Feedback Analysis

The feedback collected during testing was categorized into the following key areas:

**Usability:** Most participants found the application simple and easy to use. The structured input form, dropdown selections, and slider controls made data entry straightforward. Users appreciated that the system required minimal effort to get results.

**Accuracy Perception:** Participants felt that the predicted price values were reasonable and close to expected market prices. The inclusion of a price range increased trust in the system by indicating prediction reliability.

**Interface Design:** The modern dark-themed interface with highlighted elements was visually appealing and engaging. Users liked the clear layout and the way important elements such as the prediction button and results were emphasized.

**Speed and Performance:** The application provided instant results, which was highly appreciated by users. The fast response time improved overall user satisfaction.

**Understanding of Result:** Users were able to easily understand the predicted price output. Additional features like insights and visual indicators helped in interpreting the results better.

### 7.3 Improvements Based on Feedback

Based on the testing feedback, the following improvements were identified for future enhancement of the AutoVal-AI system:

- Add comparison with real market listings for better validation
- Include more advanced features such as location-based pricing
- Provide detailed explanation of how the price is calculated
- Improve mobile responsiveness for smaller screen devices
- Add graphical charts for better visualization of price trends
- Include user login to save previous predictions
- Expand dataset to improve prediction accuracy further

# 8. CONCLUSION

The Design Thinking process provided a structured and effective framework for addressing the challenges associated with used car price estimation. By adopting a user-centered approach throughout the project, our team was able to develop **AutoVal-AI**—an AI-powered car price prediction system that simplifies and improves the process of determining a vehicle's market value.

Through the Empathize stage, we gained valuable insights into the difficulties faced by users in the used car market, including lack of pricing transparency, dependence on dealers, and uncertainty in decision-making. The Define stage helped us clearly identify the need for a reliable, data-driven solution. During the Ideate stage, we explored multiple approaches and finalized a Machine Learning-based system as the most effective solution. The Prototype stage enabled us to transform our ideas into a functional web application, while the Testing stage validated the usability, accuracy, and performance of the system.

AutoVal-AI integrates key features such as real-time price prediction, analysis based on multiple factors (company, model, year, fuel type, and kilometres driven), confidence-based price range estimation, and additional insights like depreciation trends and vehicle condition evaluation. These features collectively provide users with a comprehensive understanding of car valuation, enabling them to make informed decisions while buying or selling vehicles.

The application emphasizes simplicity, speed, and usability, ensuring that users with varying levels of technical knowledge can easily interact with the system. By reducing reliance on intermediaries and providing transparent pricing, AutoVal-AI contributes to a more efficient and trustworthy used car market.

This project has reinforced the importance of combining technology with user-centered design principles. It demonstrates that Machine Learning, when applied thoughtfully, can solve real-world problems and enhance decision-making processes. The Design Thinking methodology helped us understand that successful solutions are built not just on technical capabilities, but on a deep understanding of user needs and practical challenges.

In the future, AutoVal-AI can be further enhanced by integrating larger datasets, advanced Machine Learning models, and additional features such as location-based pricing, real-time market comparison, and user account management. With continuous improvement and real-world deployment, AutoVal-AI has the potential to become a powerful and reliable tool in the used car ecosystem.

# 9. REFERENCES


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